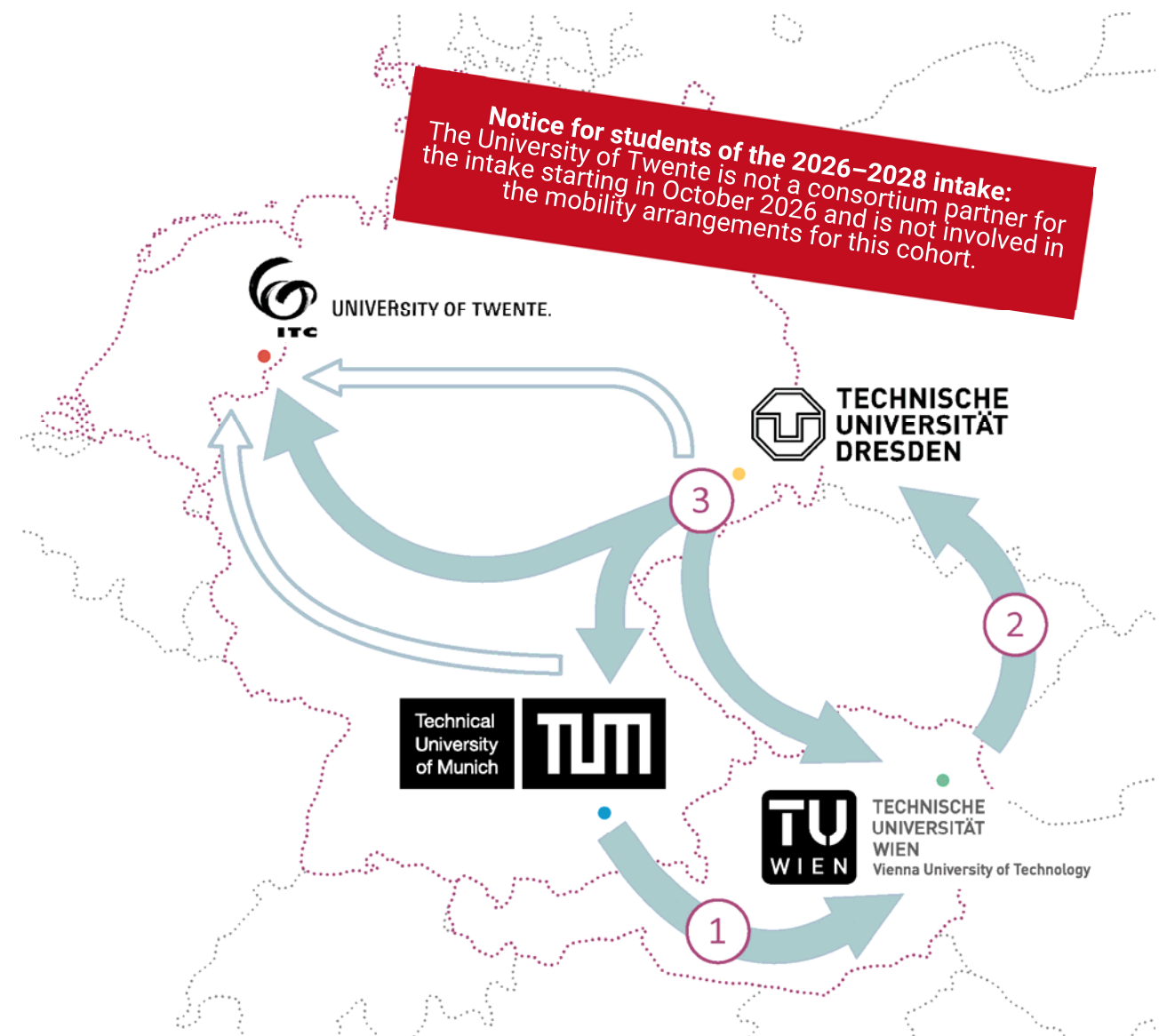


Last update: June 2026



Cartography M.Sc.

Student Handbook

Erasmus Mundus Master of Science in Cartography:
Mapping for a Sustainable World

A cooperation of:

Technische Universität München, Germany (TUM)

Technische Universität Wien, Austria (TUW)

Technische Universität Dresden, Germany (TUD)

University of Twente, The Netherlands (UT-ITC)



Co-funded by the
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Foreword

Dear students,

This handbook aims to provide you with detailed information on regulations regarding the joint Master programme in Cartography at the Technische Universität München (TUM) in Germany, at the Technische Universität Wien (TUW) in Austria, at the Technische Universität Dresden (TUD) in Germany and at the University of Twente (UT) in The Netherlands.

For this purpose, this handbook shortly outlines the major objectives and course structure of the programme as well as present regulations with respect to the different modules and the Master's thesis.

In the annexes we provide information on the specific content of each module, its scheduled dates, involved lecturers and other relevant details such as the workload of each module and the ECTS (European Credit Transfer System) points assigned to them.

We sincerely wish you a successful academic development and all the best. We are looking forward welcoming you soon!

On behalf of the Cartography M.Sc. consortium,

Juliane Cron
Programme Coordinator

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1 Introduction

Have you ever met someone who doesn't like maps or doesn't use them? Probably not. But, how many good maps do you see in daily life? Who creates recipes for mapping ever-growing data from the ever-changing environment?

The unique Cartography Master programme is dedicated to educate international talents who are able to create recipes and good maps. The programme is a professional academic approach on geodata integration, visualization and participatory user experience to foster knowledge discovery from maps. Students in the Cartography programme learn how to develop and evaluate cartographic tools on the basis of firmly established theories and methods. The focus is on developing and applying scientific methods and techniques to improve geo-information services for a diverse range of heterogeneous users. Another added value of the programme is its educational execution in locations across Europe, a historic centre of excellence in the field of Cartography. Excellently educated students from this programme filling positions not only in the cartographic research community and geosciences, but also in other related research fields that address global challenges as defined by bodies like the United Nations or the European Union.

Partners

The international Master of Science in Cartography was established in 2011 and is a joint programme of:

Chair of Cartography and Visual Analytics

Technische Universität München (TUM)
Department of Aerospace and Geodesy

Research Unit Cartography

Technische Universität Wien (TUW)
Department of Geodesy and Geoinformation

Institute of Cartography

Technische Universität Dresden (TUD)
Faculty of Environmental Sciences, Department of Geosciences

Department of Geo-Information Processing

University of Twente (UT)
Faculty of Geo-Information Science and Earth Observation ITC

Students starting the programme in October 2026 should note that the University of Twente will not be part of the consortium and will not be involved in the mobility arrangements for this intake.

All partner sites represent centres of excellence in the field of cartography, recognized at the European and global levels.

The Chair of Cartography and Visual Analytics of TUM, the Research Division Cartography of TUW, the Institute of Cartography of TUD and the Faculty of Geo-Information Science and Earth Observation (ITC) of UT complement each other in an ideal way to provide an exceptional programme with attractive job employment prospects in this interdisciplinary field.

The composition is a complementary approach of the partners' different research and educational profiles in an area, where each partner is a leader in their own country and internationally recognized institution.

2 Programme structure, mobility and content

Programme structure

The Master degree in Cartography will be achieved over two years through the allocation of 120 ECTS. Students have to obtain a total of 30 ECTS per semester. The language of tuition is English. The programme starts every year on the first day of the winter semester.

The qualification finally awarded to the students is the joint degree of all four universities "Master of Science (M.Sc.) in Cartography".

Mobility

The mobility combination has been carefully designed to eliminate overlaps of course contents, to guarantee the inclusion of necessary modules and to have the students finish within two years.

Students stay together as a group during the first three semesters of the programme. They spend their 1st semester at TUM, their 2nd semester at TUW and their 3rd semester at TUD. During the 4th semester students prepare their Master's thesis at one of the partner universities and are supervised by a staff member of that university. In addition, there will be at least one reviewer from another partner university.

In case a student cannot pass over directly after semester 1 (TUM) to semester 2 (TUW), the student does not have to interrupt the programme for one full year. In that case, the student can move in the following start of the winter semester to TUD and catch up the semester at TUW afterwards.

Thus in reasonable cases it is possible that a student studies as follows:

- TUM (winter semester),
- Break (summer semester),
- TUD (winter semester),
- TUW (summer semester),
- Master's thesis (winter semester).

In any other cases a student cannot directly pass over to the next university – the student has to wait for one year.

Programme content

The programme focuses on a comprehensive education in cartography and geoinformatics, in particular on spatial information technologies, mathematic principles of map making, spatial data handling, theories and technologies of visual analytics, geovisualization and communication of spatial information. The curriculum is structured in complementary and consecutive modules. It is based on lectures, exercises, seminars and project works.

First semester at TUM

The first semester offers different learning paths, based on the previous know-how and the additional learning requirements of the students. Thus, this semester harmonizes the student's knowledge. The basic cartographic principles will be introduced and extended and the acquired skills will form a basis for the continuing programme. Teaching during the first semester includes cartographic foundations, geoinformation, geovisual analytics, image analysis for mapping. Additionally, soft skill competences like project management skills are provided.

Second semester at TUW

The second semester contains advanced courses in cartography and geoinformatics. Basics taught in the first semester will be deepened and enhanced, especially in web mapping, location-based services and cartographic interfaces. A broad training in multimedia techniques as well as deep

knowledge of relevant programming technologies is given. Project management and presentation skills and other soft skills are trained in an integrated manner in most courses. Practical group work as well as independent projects will further improve the students' social skills.

Third semester at TUD

The third semester is dedicated to thematic specialisations in the field of cartography. Students have the opportunity to specialise in mobile cartography, remote sensing cartography and GIS related applications. An Alpine Cartography Field School will demonstrate that despite of all digital earth monitoring a direct sensual contact to a landscape is still required in much of the geo-scientific and cartographic work. From numerous research and application orientation projects complete work chains can be illustrated, which guide the student from the data capture (in the field and from remote sensing imagery) over various steps of data harmonisation, integration and other forms of value-adding to final quality 2D and 3D visualisations. Furthermore, students get insight in the research and the development work related to automated cartographic generalization and the utilization of multiple representation databases.

Fourth semester at one of the four partner universities

The fourth semester is dedicated to the Master's thesis. A large variety of Master's thesis subjects in fundamental and applied research areas in cartography will be announced. Research topics are proposed by all members of the consortium. Associate partners and the students themselves may also propose topics, which may arise, for example during an internship. Master's theses are written, presented and discussed in English. Students are required to give a mid-term presentation and the final thesis with its defence will be evaluated by a thesis assessment board.

The modules scope and credits summary are listed in Annex 1. A detailed description of each of the compulsory and elective modules is provided in the Module Handbook.

3 Starting dates and duration of the course

The Cartography Master programme officially starts on 1st of October of every year and lasts 2 years, 4 semesters, 120 ECTS.

The official kick-off of the new intake will take place during a welcoming week. The welcoming week takes place one week before the official lecture period starts. The first official lecture day at TUM in winter term 2026 is the 12th of October.

Semester organisation

Although the semester timetables differ slightly in each institution, every effort will be made to ensure that students attending the programme at different locations will start and finish in the same time period allowing adequate vacation time between the four semesters. Students have to obtain at least 30 credits (ECTS) in each semester.

Important semester dates

1st semester at TUM [1]	01.10.2026 – 28.02.2027
	Lecture period: 12.10.2026 – 05.02.2027 (exams last until end of February)
2nd semester at TUW [2]	01.03.2027 – 30.09.2027
	Lecture period: 01.03.2027 – 30.06.2027 (exams last until end of June)
3rd semester at TUD [3]	01.10.2027 – 31.03.2028
	Expected date of the 10 days Alpine Cartography Field School: "Georelief and Cartography" in the mountains close to Salzburg: end of September 2027.
4th semester at one of the partner universities	Lecture period: 11.10.2027 – 05.02.2028 (exams last until end of March)
	01.04.2028 – 30.09.2028 Duration depends on the time to finish the Master's thesis (approx. 6 months)

More information about semester dates can be found under:

[1] <https://www.tum.de/en/studies/application/application-info-portal/dates-periods-and-deadlines>

[2] <https://www.tuwien.at/en/studies/admission/academic-calendar>

[3] https://tu-dresden.de/studium/im-studium/studienorganisation/studienjahresablauf/?set_language=en

4 Admission and enrolment

The whole admission and enrolment procedure of the Cartography Master programme is coordinated by TUM. Therefore, students are registered at the admitting institution TUM for the whole time of their studies. Students are associated with the admitting and one host institution during the second and third semester. In the fourth semester, students are registered at TUM and the university employing the first supervisor of the Master's thesis. Students are registered at TUD as side Students during the first and second semester and as full students in the third semester.

Admission letter

If accepted to the programme, you can print out an acceptance letter via TUMonline. This letter is considered official, as the German version is the only legally binding letter. This also means you will not receive a hardcopy of your admission letter via regular mail.

Therefore, you should print the acceptance letter and enclose it to your student visa application for the German embassy. If your local German embassy requires an admission letter with a TUM seal or stamp, please inform us immediately and request a letter by e-mail to studium@tum.de or info@cartographymaster.eu! Otherwise you may have a long delay and ultimately don't receive your student visa on time!

Enrolment

Once all your required documents for enrolment (e.g., university degree) are received at the TUM admission office ("Immatrikulationsamt") and the semester fee is paid, you will be automatically enrolled into the TUM as a student and you will be able to print out your enrolment confirmation via your TUMonline account (as described above). There is therefore no official "enrolment day" at TUM.

Initial enrolment payment at TUM

Please pay your semester fees in full as soon as possible after notification of admission. The payment deadline is 15th of September 2026. Only after your student fees have been paid in full you are officially enrolled as a regular student at TUM. Once you are officially enrolled, you will be able to print your student documents (enrolment verification and student progress certificate) or register for courses and exams.

5 Semester and tuition fees

The total fee for the programme varies, depending on the chosen mobility (where the student writes the Master's thesis) and the country of origin. The TUM fee (or a solidary contribution to this fee) needs to be paid every semester. For more details, please see <http://cartographymaster.eu/financial-information/>. Please note: The fees are subject to change slightly every semester.

1st semester TUM:

EUR 97.00

2nd semester TUW

For non-EU nationals: EUR 726.72 tuition fee, for EU nationals: EUR 25.20 student union fee

+ EUR 97.00 TUM fee

3rd semester TUD

EUR 338.00

+ EUR 97.00 TUM fee

4th semester (varies)

The costs for the fourth semester of the MSc Cartography depend on the university where the student completes the thesis, which is the university at which the first supervisor is employed. The semester costs stated above for the respective university apply accordingly. For example, students writing their thesis at TUD pay the same costs as for the third semester: EUR 338.00, plus EUR 97.00 for TUM.

Other fees:

Alpine Cartography Field School

During the third semester the elective module “Georelief and Cartography” consists of a 10 days’ excursion to the Austrian Alps (Dachsteingebirge), which is one of the programme highlights. Estimated excursion costs are about EUR 850.

6 Erasmus Mundus scholarship

The MSc Cartography is co-funded by the Erasmus+ Programme of the European Union until the end of 2027. However, the last intake eligible for Erasmus Mundus scholarship funding started in 2025. Consequently, no Erasmus Mundus scholarships are available for the next cohort.

We are nevertheless pleased to continue the programme and to welcome a new intake of self-funded students in October 2026.

7 Other funding opportunities and sponsorship

A wide variety of foundations offer scholarships to students at all levels of study and eligibility is not limited to students with outstanding marks. If you require sponsorship, you will need to apply to one of the funding organisations.

For information about other scholarships, we would like to refer you to the comprehensive scholarship database of the DAAD, the German Academic Exchange Service:

<https://www.daad.de/de/studieren-und-forschen-in-deutschland/stipendien-finden/>

Also TUM provides scholarships for international students:

www.tum.de/en/studies/fees-and-financial-aid

Further information can be found at:

<https://globalscholarships.com/>

<https://www.scholarshipportal.com/>

<https://www.studienstiftung.de/>

www.stipendiumplus.de

www.hss.de

www.european-funding-guide.eu/

8 Health insurance coverage

All students in Germany are required to have health insurance. Therefore, students need to provide proof of health insurance coverage for enrolment at the university and, in addition, for a visa and residence permit to be granted or renewed for the purpose of academic study.

From the intake starting in 2026 onwards, students are self-funded and are therefore responsible for arranging and covering their own health insurance. The programme cannot provide health insurance for this intake. However, the programme coordinator will support students by providing detailed information and guidance.

Students who are not insured in their home country must obtain health insurance coverage in Germany from the insurance provider of their choice. Possible options include:

- German public state health insurance, such as TK or Barmer, or
- Private health insurance, such as Dr. Walter or Care Concept.

When choosing an insurance provider and tariff, students should bear in mind that the insurance should also be sufficient for studying and staying in Austria.

Students who choose German public state health insurance, such as TK or Barmer, may need additional accident insurance for Austria. Another visa requirement in Austria is to provide confirmation of health insurance valid for the period of stay and covering costs of at least EUR 30,000.

Private health insurance is often less expensive than German public state health insurance. However, students should be aware that, depending on the insurance provider and tariff, they may have to pay medical costs themselves first when visiting a doctor and then request reimbursement from the insurance provider. The programme has good experience with both Dr. Walter and Care Concept.

For students considering Dr. Walter, further information is available here: <https://www.dr-walter.com/en/info-portals/german-student-insurance/>. Students who come to Germany from other European and non-European countries to study may need different insurance coverage depending on the phase of their stay. The Dr. Walter information portal <https://www.german-student-insurance.com/en/> for international students in Germany provides guidance on these questions.

For students choosing Care Concept, further information is available here: https://www.care-concept.de/private_studentenversicherung_eng.php?navilang=eng

For the second semester in Austria, information on Care Austria is available here: https://www.care-concept.de/oesterreichversicherung_eng.php?navilang=eng

Students who choose private health insurance, such as Dr. Walter or Care Concept, must provide proof for enrolment at TUM that they are privately insured. Therefore, they need to obtain a letter confirming the exemption from the compulsory German public state health insurance. The programme coordinator will assist students with this process.

More information is available on the programme website: <http://cartographymaster.eu/health-insurance/>

Students may also consult the TUM website on mandatory health insurance: <https://www.tum.de/en/studies/application/enrollment-info-portal/mandatory-health-insurance>

Please read the information on the TUM website carefully, as it does not specifically address the situation of students who study abroad for one or two semesters as part of their programme.

9 Visa and residence permit

Visa and entry regulations for Germany

Students with all European Community and EEA/EFTA (European Economic Area / European Free Trade Association and Switzerland) nationalities are exempt from this regulation. After arriving in Germany you only need to register their residential address at the residence registration office.

Visa application

Students with a nationality of the United States, Australia, Brazil, New Zealand, Japan, Canada, the Republic of Korea or Israel are usually allowed to enter Germany without a visa. After entering Germany, they must apply for a residence permit ("Antrag auf Erteilung der Aufenthaltserlaubnis") at the local foreigners' registration authority within three months of entering the country.

However, due to the currently long processing times for residence permits in Munich, we recommend that these students also check whether they can apply for a student visa before entering Germany. Please note that not every German embassy or consulate may issue a visa if entry into Germany without a visa is possible.

Students from all other countries must apply for a visa at the embassy or consulate of the Federal Republic of Germany in their home country before entering Germany. The visa should be applied for at least two months in advance. Once in Germany, students must apply for a residence permit ("Antrag auf Erteilung der Aufenthaltserlaubnis") at the local foreigners' registration authority. Non-EU citizens will receive a residence permit for the respective programme of education.

The programme coordinator can support students by issuing visa support letters and, if necessary, by contacting embassies or consulates in case of difficulties during the visa application process.

Important:

International students who have proof of admittance to a German university will be issued a so-called *student visa* ("Studentenvisum"), most probably Schengen Visa.

Please do not travel to Germany on a tourist visa! This cannot be changed into a student visa and students are not allowed to study in Germany on a tourist visa.

For general information see the website of the Federal Foreign Office: www.auswaertiges-amt.de

Register your address at the residence registration office ("Anmeldung")

After arriving in Germany you must register your residential address (street, apartment, landlord) at the residence registration office ("Bürgerbüro") within one week. Also if you move to other accommodations within Munich or move out of Munich, you must register your new address or notify the office that you are leaving Munich.

To register at the residence registration office, you will need:

- Your valid passport or identification card,
- A special registration form which can be downloaded from the Internet or is available free of charge at the registration office,
- Landlord confirmation ("Wohnungsgeberbestätigung") – this is a document you receive from your landlord when you move into or out of your room. You need to present it to the registration office when you register or deregister your address. Sometimes, authorities may also ask to see your rental agreement. Before signing any rental contract, please make sure that the landlord is able and willing to issue the landlord confirmation.

After completing the registration form you can register your stay either in person or in writing. You will receive a written confirmation of registration. This proof of registration is often required by other institutions or authorities.

Important: When leaving Munich, you must fill out a form in order to give notice of your departure!

For more information, see:

<https://stadt.muenchen.de/en/immigration.html>

Residence permit

In case you enter Germany with a visa that is shorter than 2 years, you must apply for a residence permit via the Munich Foreigners Office immediately after your arrival in Germany. A residence permit will be granted/renewed only if a corresponding application is submitted. The necessary documents are:

- An application form (completed in full) for the granting/renewal of a residence permit,
- Registration of your address in Munich,

- A valid passport,
- Proof of adequate funding,
- Proof of adequate health insurance,
- Proof of the purpose of your stay in Germany, e.g., a valid university registration,
- A biometric passport photograph.

Important:

For Non-Europeans students: During your first semester at TUM you need to apply either for a two-year-residence-permit for Germany or for a *visa for Austria for the second semester*. Therefore, your passport should value at least until mid of the 3rd semester and the passport needs to have at least 2 empty pages.

Visa and entry regulations for Austria will be announced after your arrival in Germany. If you have obtained a two-year-visa for Germany, you do not need to apply for an Austrian visa. Please have a look on some general information about visa regulations for these countries on our website <http://cartographymaster.eu/visa-information/>.

We aim to support students in obtaining a residence permit for Germany for the entire duration of the study programme. If the residence permit is issued for the full 24 months, students may not need to apply for an additional visa for Austria, provided that their stay in Austria is shorter than six months and they return to Germany within this period. This can reduce the administrative effort for students.

10 Bank account

During your stay in Europe, you will need a bank account, e.g., to handle rental payments.

Students may open a bank account with the banking provider of their choice in Germany. This may be a traditional bank with local branch offices, such as Deutsche Bank or HypoVereinsbank, or an online banking provider, such as N26 or Revolut.

Documents:

The documents you need to bring with you when you go to a bank branch office are:

- Passport,
- Registration of your residential address ("Anmeldung"),
- The letter of confirmation of the study programme.

To open an account:

Bring your documents to a bank branch and ask the staff member at the bank counter to open a bank account for you. You need to fill in an application form and usually sign some additional documents, e.g., the online banking application.

Depending on your account in the following weeks, you will receive via post, e.g., a bank card, PIN, online banking password and online banking TAN.

Note that if you have already opened a German bank account in your own country, you still need to activate it in a bank branch office upon your arrival.

The closest bank branch of Deutsche Bank to TUM is located at Barer Str. 62, 80799 München.

11 Preparing to study for the first semester at TUM

Higher education in Europe is strongly influenced by the Enlightenment achievements and Humboldtian idea of science and research; therefore, it places a large emphasis on the independent study.

As a student you will be asked to make notes, sketch, read widely on a subject and search for additional information to you find your own way of thinking and analysing the world. Key skills to succeed in the learning process are good organization and time management. With the support of lecturers and supervisors you will develop during your studies also other soft skills such as project management, social communication and language competences.

Arriving in Munich

Before your arrival in Germany, please make a few preparations to ensure that you can use your local bank account and your credit / debit card with no problems. Card acceptance varies country by country, and even city by city. When arriving in Munich, please keep in mind to have approx. EUR 200 / USD 250 (can be changed to EUR at the airport) in cash. It will be useful for your spending during first days in Munich, such as buying tickets for public transportation or food. Check with your banking provider if there will be any foreign transaction fees if you use your card in Germany and ask if your bank has international partnerships that allow you to use some ATMs for free.

Popular course types

The classification and structure of courses may differ slightly from the educational system you are familiar with from your previous university studies. The MSc Cartography programme is based on lectures, exercises, seminars and project work in the form of compulsory and elective modules. You should carefully read the requirements for elective modules and prepare yourself for the corresponding workload: 1 ECTS credit corresponds to approximately 30 hours of work.

During the Welcome Week, a special session will be dedicated to explaining the full system to you.

Course types

V ("Vorlesung")	Lecture	The most traditional form of teaching – course taught by a lecturer, often with many students in a large lecture hall. Lectures are often accompanied by relevant exercises and homework.
Ü ("Übung")	Exercise	Practical exercises connected to lectures themes.
E ("Exkursion")	Excursion	Field trips, day excursions and practical in-situ trainings.
P ("Projekt")	Project	Practical training in a subject area, usually in work groups examined by reports and presentations.

Attendance at lectures is generally not compulsory – in contrast to exercises, excursions, practical trainings and projects. However, we strongly recommend that you attend lectures regularly, as their scope containing also research findings and discussions is not fully covered by digital handouts and presentations. Most lecturers upload the lecture handouts, presentations, homework and other documents via the *Moodle e-learning portal* or *TUMonline*.

If you must miss a lecture, to go over what you missed we recommend checking the materials on the Moodle platform, being in touch with your classmates and finally meeting the lecturers during their office hours.

Computer work

Almost all of the courses subjects will require a computer work as a part of the learning process. You will be usually asked to perform analysis of geodata, prepare different maps and visualizations, type your assignments or take an online-exam.

At the university you may use computers in private study, lecture rooms and working spots. At TUM there are known as CIP-Pools and there are placed in the main campus in rooms 3209, 3238 as well as 0712 (Cartography CIP-Pool,) all situated at TUM Campus Arcisstraße 21/ City Center. You can login to the computers with your TUMonline user name and related password.

For the following semesters at the partner universities an own laptop is absolutely necessary.

Software

As a TUM member you have access to a great deal of software and hardware for free or at low cost, e.g., Microsoft-Products (via StudiSoft), Apple on Campus, ESRI (ArcGIS) and other products like ChemDraw or Origin. Usually you only need your TUM ID and password for download.

Further information for student can be found under www.it.tum.de/en/students/#c4480

Student Card

The Student Card is a multifunctional chip card that acts as the TUM student ID card. Moreover, it works as a library card and student services electronic payment card, e.g., for payment in canteen, cafeteria and snack vending machines.

You can load your card in charging stations available in all canteens and cafeterias of the TUM Student Union. The Student Card can be loaded with the following amounts EUR 5, EUR 10, EUR 20 and EUR 50 in bank notes. Another option is to use autoloan, a system for automatic top-up via direct debit.

For more information about Student Card have a look on www.tum.de/en/studies/application-and-acceptance/student-card

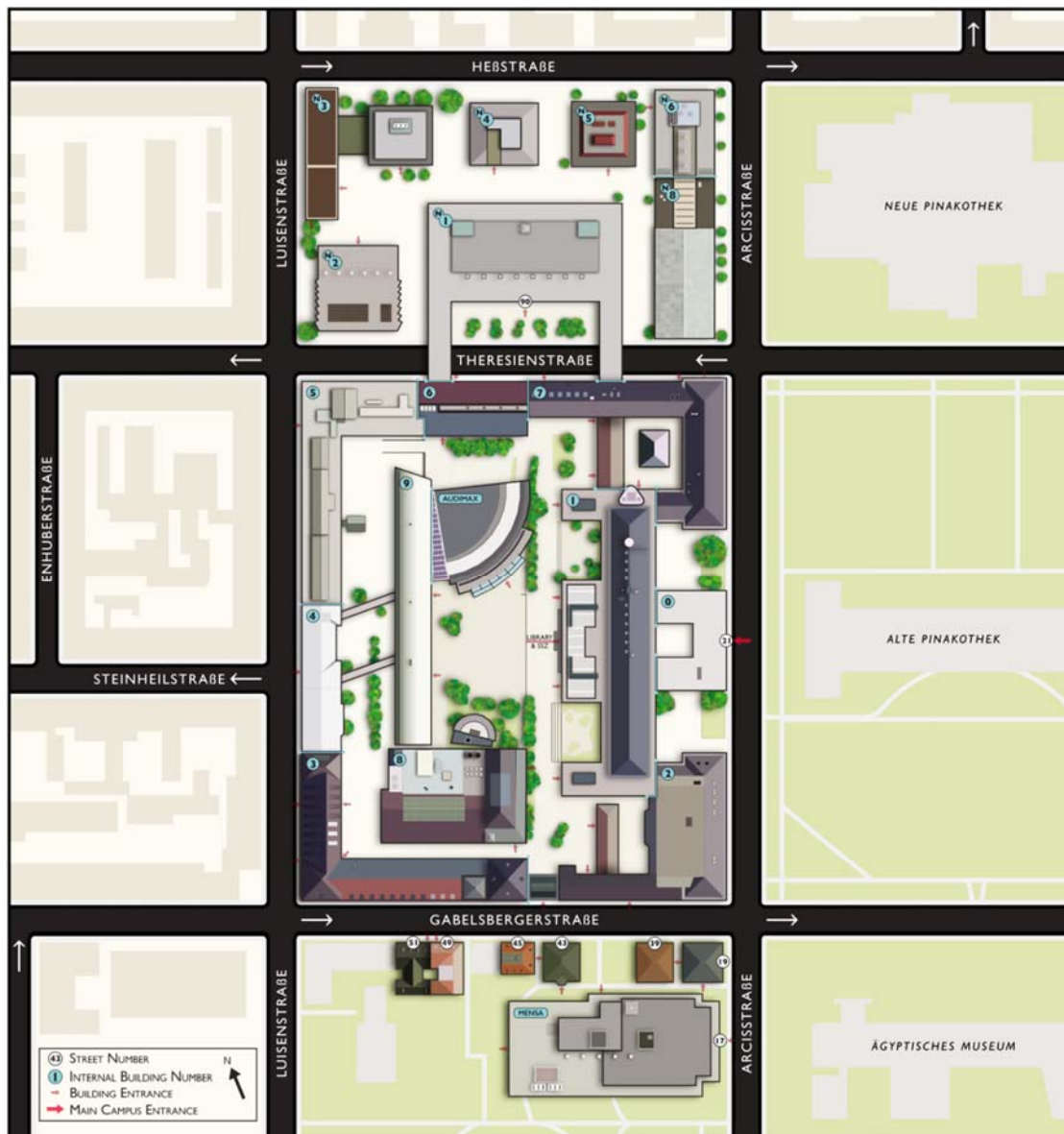
Semester ticket

The Deutschlandticket was introduced on May 1, 2023. The ticket entitles the holder to use regional public transport (including bus, subway, S-Bahn, regional trains) throughout Germany. For students enrolled in a Bavarian University, an even cheaper **43-euro-student Deutschlandticket** is available.

Have a look on <https://www.tum.de/en/studies/during-your-studies/mobility-at-campus> to find out more about 43-euro-student Deutschlandticket ticket.

Navigating

One of the most useful tools at the beginning of your stay at TUM will probably be NavigaTUM: <https://nav.tum.de/> - a portal showing the location of university rooms and buildings. Keep in mind, that sometimes not all rooms are digitalized and available at the website.



TUM Campus map created by students of Cartography Master programme.

Housing

It's not easy to find a place to live in Munich – but it's not impossible either! TUM supports students in their search for accommodation, providing personal advice, in-house listings and useful information to ensure that you can quickly find a place to call your own.

With high rents and not enough rooms in halls of residence, the Munich rental market is one of the most competitive in Germany – especially for students. More people moving to Munich every year – many of them attracted by the excellent study opportunities that Munich has to offer.

All information about finding a place to live in and around Munich can be found on <https://www.tum.de/en/university-life/accommodations/>.

12 Student life

There are various websites that can help you navigate your way and provide information on shopping, eating out and local events.

Our website www.cartographymaster.eu provides all information about student life in the cities of your studies.

Working besides studying

The study schedule allows you to work beside the course just a little bit. 30 credits are to be obtained in each semester which means a workload of approx. 900 hours.

EU/EEA students have free access to the German job market and are practically equivalent to German students. International students from other countries can work a total of 120 full or 240 half-days per calendar year, in addition to a part-time student job at a university or public institute of science. If you want to work more, you need a permit from the "Agentur für Arbeit" (Federal Employment Agency) and the foreigners' authority. The basic rule is that gainful employment must not impair the course of study! Whether you are issued a work permit largely depends also on the condition of the local job market.

The recruitment process starts generally with detailed preparation of your application documents (CV, cover letter and references or certificates) and sending them in response to a job offer or through an unsolicited application. The second part is being invited to 1– 3 rounds of job interviews (telephone, Skype or in person) or an assessment centre followed by a job offer.

For the first semester in Munich information and support about job finding is provided by [*TUM Job Börse*](#). Important information, as well as job offers for international students can be found at the [*Alumni and Career website*](#). Other offers are provided by official [*Munich City Portal*](#).

Consulting / Information sources

Our programme webpage functions as a comprehensive information source. Additionally, at each of the participating universities there are excellent services and supports for international students:

- TUM International Center: www.international.tum.de/en/home
- TUW International Office: www.ai.tuwien.ac.at/international
- TUD International Office: www.tu-dresden.de/internationales
- UT International Community: <https://www.utwente.nl/en/education/master/internationalstudents/preparations/>

13 Check list

What should I do first after I got accepted for the Cartography Master programme?

- ☐ Confirm acceptance in TUMonline
- ☐ Pay the semester fee for the first semester at TUM
- ☐ Get the student visa,
- ☐ Find an accommodation in Munich,
- ☐ Register the new address,
- ☐ Open a bank account,
- ☐ Get a student card,
- ☐ Possibly sign-up for the Deutschlandticket,
- ☐ Sign up for university facilities (e.g., library, lectures, moodle via TUMonline),
- ☐ Apply for a residence permit directly after arrival (non-EU students).

14 Contact information

Programme coordinator and Local coordinator at TUM:

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15 The Erasmus Mundus Students and Alumni Association (EMA)

Over 15.000 bright international students took part in the Erasmus Mundus programme already. Almost 9.000 of them are members of the EMA association!

EMA offers an international professional and personal network for Erasmus Mundus students and graduates. The members can join professional networks and service teams to network and become active within the Association. Jobs board and special entrepreneurship sections are also offered to the members.

EMA members are ambassadors of the Erasmus Mundus, share the idea of international education and spread the messages around the world. Since its foundation EMA members have initiated or taken part in over 150 fairs, information seminars and promotional activities worldwide.

EMA serves as a channel of communication for students, alumni, universities and the European Commission and responds to the suggestions of the students through Course Quality Advisory Board.

Programme Representatives and Country Representatives EMA enable the potential Erasmus Mundus applicants to get in touch with the current students and graduates to receive a piece of advice and recommendations based on the experience of previous generations.

More information: www.em-a.eu/en/home.html

16 Annexes

The curriculum table provided in this annex reflects the current planning status and is subject to change until 15 September. Adjustments may be necessary, for example, if individual courses cannot be offered as planned. Please note that the module handbook contains additional module descriptions beyond the modules offered in the upcoming semester.

Annex 1: Curriculum for the intake 2026-2028

Nr.	Module name	Type	SWS	ECTS	Exam	Exam duration
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1st semester at TUM (30 credits):

Compulsory modules (22 credits in total):						
BV300025	Cartographic Foundations	VÜ	3	5	sch.	90 min
BV300003	Geo-Information	VÜ	4	6	sch.	60 min
ED110175	Geovisual Analytics	VÜ	4	5	sch.	90 min
ED110091	Image Analysis for Mapping	VÜ	4	6	sch.	90 min
Elective modules (minimum 8 credits):						
BV470015	Geodatabases	VÜ	3	5	sch.	60 min
ED110031	Geostatistics and Geomarketing	VÜ	4	5	sch.	90 min
BGU30046	Mapping Project	P	3	5	P.	
BV030012	Engineering Databases	V	2	3	sch.	60 min
LRG1500	Principles of Spatial Data Mining and Machine Learning	VÜ	3	3	sch.	60 min
LRG1501	Selected Topics in Big Geospatial Data	VS	3	3	R.	
SZXXXX	English (please see full list online)	V	2	3	m.	30 min

2nd semester at TUW (30 credits):

Compulsory modules (30 credits in total):						
BV300027	Cartographic Theories and Applications	VÜ	6	9	sch.	120 min
BV300028	LBS and Multimedia Cartography	VÜ	7	10	sch. (30 %) + P. (70 %)	120 min
BV300029	Cartographic Publishing	VÜ	4	5	sch.	120 min
BV300030	Applied Cartographic Research and Development	VP	5	6	P.	

3rd semester at TUD (30 credits):

Elective modules (minimum 30 credits in total):						
BV300033	Georelief and Cartography - Morphogenetic and Environmental Understanding	VE	8	10	m.	30 min
ED110245	Affective and Multi-Perspective Map Communication	VÜ	3	5	m. (50 %) + P. (50 %)	20 min
ED110247	Foundations of Virtual Reality	VÜ	3	5	sch.	120 min
BV300031	Mobile Cartography	VÜ	4	10	sch. (40 %) + P. (60 %)	60 min
ED110246	Topographic Map Production	VÜ	4	5	sch. (50 %) + P. (50 %).	90 min
BV300037	Remote-Sensing-based Environmental Mapping	V	2	5	m.	20 min
BGU30051	Laser Scanning and DTM Generation	V	3	5	sch.	90 min
BGU30050	Geodata Infrastructures	V	3	5	sch.	90 min

4th semester either at TUM, TUW, or TUD (30 credits in total):

BGU0MTC A15	Master's Thesis	P		30	Wiss. Ausarb. (80%) + m. (20%)
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Shortcuts meaning

SWS = Credit hours

V = Lecture

Ü = Exercise

S = Seminar

E = Excursion

P = Project

R = Report (incl. Presentation)

sch. = written

m. = oral

wiss. Ausarb. = written Report

Annex 2: Module handbook

Please note that the module handbook contains descriptions of all approved modules. Not all modules listed are necessarily offered in each semester or academic year. The actual course offer is reflected in the curriculum table and may be subject to change.

www.cartographymaster.eu/wp-content/documents/CARTOGRAPHY_Module_Handbook.pdf